

AN INTERNSHIP -2 - PROJECT REPORT
on
MACHINE LEARNING USING BY PYTHON

As a part of
Internship Credit (CE0318)

Submitted by

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ABSTRACT - the internship project focused on 'machine learning with python' and covered various aspects of python programming, data handling, and machine learning.

Starting with python basics, the internship explored advanced

Libraries such as NumPy, pandas, and matplotlib, followed by

Implementation of algorithms using scikit-learn. The

Final project involved classifying customers as smokers or

Nonsmokers using logistic regression. The internship provided

Strong

Foundations in programming, data science, and

Practical implementation of machine learning models.

TECHNOLOGIES USED - python 3.10 - numpy - pandas - matplotlib - scikit-learn - seaborn - visual studio code

SUMMARY OF WORK

- Learned python basics: environment setup, syntax, variables, loops, functions, and Conditionals.
- Explored core data structures: list, tuple, set, and dictionary.
- Practiced numerical operations with numpy and data handling with pandas.
- Created visualizations using matplotlib.
- Implemented machine learning models (knn, regression, kmeans) with Scikit-learn and evaluated results.

Tasks Assigned by industry mentor

Numpy Tasks:

Task 1: Array Creation and Reshaping

Create a Numpy array containing numbers from 1 to 20. Reshape the array into a 4x5 matrix. Flatten the reshaped array back to a 1D array.

Task 2: Indexing, Slicing, and Conditional Selection

Create a 5x5 Numpy array with numbers from 1 to 25. Extract the second row from the array. Extract the last column from the array. Find and display all elements in the array that are greater than 15.

Task 3: Mathematical and Statistical Operations

Generate an array of 10 random integers between 10 and 100. Find the maximum, minimum, mean, sum, and standard deviation of the array. Sort the array in ascending order.

Task 4: Arithmetic Operations and Universal Functions

Create two Numpy arrays: [1, 2, 3, 4] and [10, 20, 30, 40]. Perform element-wise addition, subtraction, multiplication, and division. then Apply square root and exponential functions to one of the arrays.

Task 5: Special Arrays and Concatenation

Create a 3x3 array of all zeros. Create a 3x3 array of all ones. Create a 3x3 identity matrix. Concatenate the zero and one matrices horizontally.

Code Example (Numpy)

```
import numpy as np

# 1. Array creation
a = np.array([1, 2, 3])
b = np.array([[1, 2], [3, 4]])

zeros = np.zeros((2, 3))
ones = np.ones((2, 2))
eye = np.eye(3)
full = np.full(shape=(2, 2), fill_value=7)

arange = np.arange(0, 10, 2)
linspace = np.linspace(start=0, stop=1, num=5) # 5 values between 0 and 1
rand = np.random.rand(2, 2) #positive float value
randn = np.random.randn(2, 2) #float value positive negative both
randint = np.random.randint(low=0, high=10, size=(2, 2)) #integer value

# print(a)
# print(b)
# print(zeros)
# print(ones)
# print(eye)
# print(full)
# print(arange)
# print(linspace)
# print(rand)
# print(randn)
# print(randint)
```

Output (Numpy)

```
[1 2 3 4 5 6 7]
[[1 2 3]
 [4 5 6]]
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
[[1. 1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1. 1.]]
[[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]
[[6 6 6 6]
 [6 6 6 6]
 [6 6 6 6]
 [6 6 6 6]]
```

```
1  import pandas as pd
2  import numpy as np
3
4  # SERIES CREATION
5
6  #1. From list
7  a1 = pd.Series([10, 20, 30])
8  print("Series from list:\n",a1)
9  # 2. From tuple
10 a2 = pd.Series ((1, 2, 3))
11 print("\nSeries from tuple:\n",a2)
12 # 3. From numpy array
13 a3 = pd.Series (np.array([100, 200, 300]))
14 print("\nSeries from numpy array:\n",a3)
15 # 4. From dictionary
16 a4 = pd.Series({'a': 1, 'b': 2, 'c': 3})
17 print("\nSeries from dictionary:\n",a4)
18 #5. With custom index
19
20 s5 = pd.Series( data=[5, 10, 15], index=['x', 'y', 'z'])
21
22 # print("\nSeries with custom index:\n", s5)
23 ## 6. Scalar value
24
25 #s6 = pd.Series (5, index=['a', 'b', 'c'])
```

```
1 import matplotlib.pyplot as plt
2 import numpy as np
3
4 # Sample data
5 x = [1, 2, 3, 4, 5]
6 y = [10, 15, 13, 18, 16]
7
8 # Histogram data
9 hist_data = [12, 15, 13, 18, 16, 12, 15, 19, 20, 14, 17, 16]
10
11 # Pie chart data
12 pie_labels = ['A', 'B', 'C', 'D']
13 pie_sizes = [25, 35, 20, 20]
14
15 # Create subplots (3 rows, 2 columns)
16 fig, axes = plt.subplots(nrows=3, ncols=2, figsize=(12, 10))
17 fig.suptitle("Different Types of Graphs using Matplotlib", fontsize=16)
18
19 # ----- Line Plot -----
20 axes[0, 0].plot(x, y, color='blue', marker='o', linestyle='--')
21 axes[0, 0].set_title("Line Graph")
22 axes[0, 0].set_xlabel("X-axis")
23 axes[0, 0].set_ylabel("Y-axis")
24 axes[0, 0].grid(True)
25
```